AETHER MK-1: Ambient Edge-Computing Topology for Non-Line-Of-Sight Human Emergency Rescue

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Abstract—During structural collapses, the critical window to locate trapped survivors is 72 hours. Traditional Search and Rescue (SAR) technologies exhibit severe operational limitations: thermal cameras fail to penetrate solid concrete, rescue dogs are impeded by environmental hazards, and acoustic microphones require the victim to be conscious and vocal. This paper presents AETHER MK-1, a novel, ultra-low-cost, non-line-of-sight human detection radar system utilizing standard 2.4GHz Wi-Fi signals. By deploying a localized Wi-Fi network over a debris field, AETHER leverages a TinyML neural network deployed on a commodity ESP32 microcontroller to analyze the multipath interference (Received Signal Strength Indicator or RSSI fluctuations) caused by the physical movements of trapped bodies. The system operates completely offline via edge computing, functioning as an ultra-fast binary classifier (Dead Rubble vs. Trapped Survivor). Experimental results demonstrate high reliability in detecting human motion through non-metallic barriers, providing immediate proof of life for rescue teams.

***Index Terms*—**Edge Computing, Human Detection, Search and Rescue, TinyML, Wi-Fi Sensing, Machine Learning, ESP32.

I. INTRODUCTION

In the aftermath of structural collapses caused by earthquakes or industrial accidents, the rapid localization of survivors is the paramount objective of emergency response teams. Medical statistics dictate a 72-hour critical survival window, after which mortality rates increase catastrophically. However, locating individuals buried under feet of concrete, steel, and rubble presents immense physical challenges that current multi-million-rupee technologies struggle to overcome.

Current SAR systems rely almost entirely on either direct line-of-sight (optics) or victim cooperation (acoustics). When a building collapses, void spaces are filled with thick dust and smoke, blinding camera robots. Solid concrete walls block infrared heat signatures, rendering thermal imaging useless. Furthermore, unconscious or severely injured victims cannot scream or tap on pipes, rendering acoustic microphones ineffective. There is a critical need for a low-cost, mass-deployable technology capable of finding human life where traditional sensors are blind.

To address these life-threatening limitations, we propose AETHER MK-1: an Ambient Edge-Computing Topology for Human Emergency Rescue. By pivoting ultra-low-cost Wi-Fi telemetry away from domestic automation and into tactical disaster response, AETHER utilizes 2.4GHz electromagnetic propagation to penetrate structural barriers. A dedicated on-device neural network analyzes the RF scattering caused by a human body, identifying proof of life entirely without optical visibility or human consciousness.

II. RELATED WORK AND LIMITATIONS

The domain of human detection relies heavily on mechanical and optical solutions. Traditional Search and Rescue Robots utilize ESP32-CAM modules or Passive Infrared (PIR) sensors. While effective in open areas, cameras fail in non-line-of-sight environments. PIR sensors detect thermal radiation, which is fundamentally incapable of penetrating solid concrete.

In the realm of RF sensing, Smart Home Wi-Fi Sensors (such as ESPectre) utilize microcontrollers to detect human presence for home automation. However, these systems require a clean, static baseline of an empty room to function and fail completely in chaotic, debris-filled environments. Conversely, Advanced CSI Trackers (like RuView and Wi-Fi DensePose) extract complex Channel State Information to map skeletal movement through walls. While highly accurate, these systems are computationally expensive, fragile, and too sensitive to environmental drift for rapid disaster deployment.

III. SYSTEM ARCHITECTURE

The AETHER MK-1 system is engineered around an ultra-low-cost, two-node hardware architecture utilizing the DOIT ESP32 DevKit V1, allowing mass deployment across disaster zones for under ₹1,000 per node.

A. The Transmitter Node (Illuminator)

The TX Node functions as an active RF illuminator. Configured in Access Point mode, it continuously broadcasts a customized 2.4GHz Wi-Fi grid (SSID: AETHER_TX_MK-1). Powered independently via a standard 5V power bank, it requires no external wiring and serves solely to saturate the targeted debris field with electromagnetic radiation.

B. The Receiver Node (AI Brain & Radar)

The RX Node is responsible for data acquisition and neural network inference. Instead of utilizing traditional slow network scanning, the RX Node utilizes a "Hard Lock Architecture," directly connecting to the Illuminator's network to prevent signal dropout. It continuously polls the Received Signal Strength Indicator (RSSI) via the WiFi.RSSI() function at an interval of 8 milliseconds.

C. Alert Mechanism

An integrated alarm circuit is connected to the RX Node. A 5mm Red LED and a 330Ω current-limiting resistor are wired to a safe GPIO pin (Pin 4). When the on-device AI classifies a positive human detection, the microcontroller drives the pin HIGH, triggering a physical visual alert that does not require an external graphical user interface or cloud connectivity.

IV. METHODOLOGY AND MACHINE LEARNING PIPELINE

The core intelligence of AETHER MK-1 is driven by a TinyML binary classifier, optimized via the Edge Impulse platform to run entirely offline on the ESP32's limited memory footprint.

A. Data Acquisition and Feature Extraction

Human movement introduces multipath interference, altering the amplitude and phase of propagating RF waves. A custom Python script logged raw RSSI telemetry across an active serial connection to build the foundation datasets: "real_idle.csv" (static environment, dead rubble) and "real_struggle.csv" (environment containing active human movement). The neural network aggregates this time-series data into a 125-sample memory buffer, representing exactly one second of physical environment analysis.

B. Neural Network Inference

The raw RSSI data undergoes Digital Signal Processing (DSP) to extract spectral features before being fed into a quantized neural network. The model classifies the 125-sample buffer array and outputs confidence percentages for the two states. The entire C++ inference library (AETHER_MK-1_inferencing.h) is compiled directly into the Arduino IDE framework. If the "struggle" confidence interval exceeds the 60% threshold, the system flags a human detection event.

V. EXPERIMENTAL SETUP AND RESULTS

To validate the RF-based human sensing approach, the AETHER MK-1 prototype was tested under rigorous environmental conditions simulating non-line-of-sight obstruction.

1. Static Control (Empty Room): The system maintained an "idle" confidence of 91.3%, successfully ignoring ambient baseline noise.
2. Dynamic Target (Walking behind wall): Human movement behind an opaque barrier triggered a "struggle" classification at 87.2% confidence.
3. Micro-Movement (Hand waving behind concrete): Smaller scale movement behind dense structural material registered a 79.8% detection confidence, instantly triggering the physical GPIO4 Red Alert LED.

The system demonstrated a 100% recovery rate from artificial network disconnections, re-establishing the Hard Lock to the TX node within 2.1 seconds. During a continuous 30-minute stress test, the inference engine remained stable with zero memory overflow crashes, confirming the viability of the C++ Edge Impulse deployment on commodity hardware.

VI. LIMITATIONS AND FUTURE WORK (MK-2 ROADMAP)

While MK-1 successfully proves the underlying physics of RF detection, it is restricted by its reliance on RSSI volume. In Test Case 04 (Human breathing only, completely motionless), the system failed to trigger the alarm (71.3% idle confidence). RSSI requires active physical movement; it cannot reliably detect a fully unconscious, immobile victim.

The AETHER MK-2 development roadmap addresses this by transitioning from RSSI to Channel State Information (CSI) phase-shift extraction. CSI provides subcarrier-level hyper-sensitivity, capable of detecting the millimeter-level expansion of a human chest cavity. Furthermore, MK-2 will transition to dual ESP32-S3 processors, incorporate IP67-rated shockproof custom PCBs, and utilize an asynchronous

web server for mobile command deployment, making the system field-ready for the National Disaster Response Force (NDRF).

VII. CONCLUSION

AETHER MK-1 successfully demonstrates that non-invasive human detection through solid structural barriers is achievable using standard, low-cost Wi-Fi hardware. By leveraging an on-device TinyML neural network, the system translates chaotic multipath interference into an immediate, life-saving binary output. Built for under ₹1,000 per node with zero reliance on cloud computing or cellular infrastructure, AETHER proves that highly scalable edge computing can solve the critical limitations of traditional thermal and optical rescue technology—effectively finding human life where eyes cannot see.

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